1. Prediction inputs

Dementia Risk Predictor

Healthcare Analytics · CA-III · Educational prototype

Explore how brain-scan measurements and clinical details influence a machine-learning estimate for the OASIS study groups.

For learning only. This prototype is not a medical tool and must not be used for diagnosis or treatment decisions.

Dashboard section

Predict | How it works | Results and limits

Prediction

Choose an example, adjust the inputs, then update the estimate.

Typical nondemented person | Typical demented person | Low MMSE, small brain volume

Brain scan measurements

eTIV (estimated total intracranial volume) 1469

nWBV (normalized whole brain volume) 0.726

Derived ASF: 1.195

Patient details

Age 76

Sex: Female

Socioeconomic status (1 = highest, 5 = lowest) 2

MMSE score (out of 30) 28

Update estimate

Estimate

2. SHAP explanation

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Dashboard section

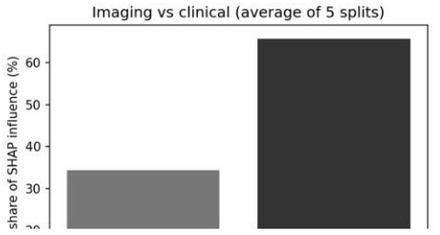
Predict | How it works | Results and limits

How the estimate is produced

A Random Forest trained on OASIS-2 combines three scan measurements (eTIV, nWBV, and ASF) with age, sex, socioeconomic status, and MMSE. SHAP summarizes how the fitted model used those inputs for an estimate.

Relative influence by input group

Across five analysis splits, imaging accounts for about 34% of mean absolute SHAP influence and clinical details about 66%.



3. Results and limits

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Evaluation results

OASIS-2 contains 150 people and 373 visits. All visits from each person were kept together during cross-validation. The OASIS-1 results below are a separate external test.

Cross-validation on OASIS-2: modality analysis

Feature set	Mean ROC-AUC	Lowest seed	Highest seed
Imaging only	0.639	0.619	0.655
Clinical only	0.742	0.739	0.744
Combined	0.793	0.777	0.817

Cross-validation on OASIS-2: classifier comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	AUC lowest seed	AUC highest seed
Logistic Regression	0.777	0.813	0.708	0.757	0.851	0.843	0.858
Random Forest	0.729	0.739	0.692	0.714	0.786	0.77	0.811
SVM	0.726	0.74	0.682	0.709	0.781	0.764	0.796

External test: trained on OASIS-2, tested on OASIS-1

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.808	0.816	0.8	0.808	0.896